



EduRev

Important Formulas & Points to Remember: Geometric Twins

Table of Contents

1. Meaning of Geometric Twins
2. How to Identify Congruent Figures
3. Congruence of Triangles
4. Important Properties Derived From Congruence
5. How to Check Congruence for Common Shapes

[View More](#) 

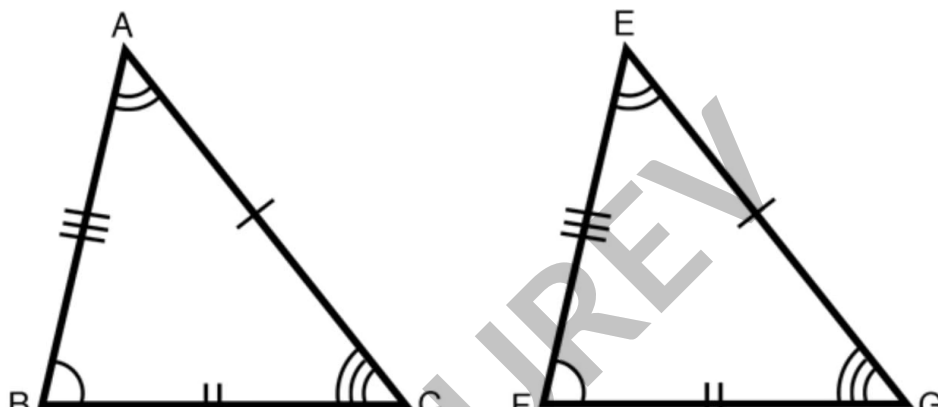
1. Meaning of Geometric Twins

Figures that have **the same shape and the same size** are called **congruent figures**.

Congruent figures can be placed on top of each other so that they **fit exactly**.

A figure may be **rotated, flipped, or moved**, and it will still be congruent as long as its shape and size remain unchanged.

Congruent Triangles



$$\triangle ABC \cong \triangle EFG$$

2. How to Identify Congruent Figures

Two figures are congruent if:

- Their **corresponding sides** are equal.
- Their **corresponding angles** are equal.
- They can be **superimposed** perfectly.

Examples:

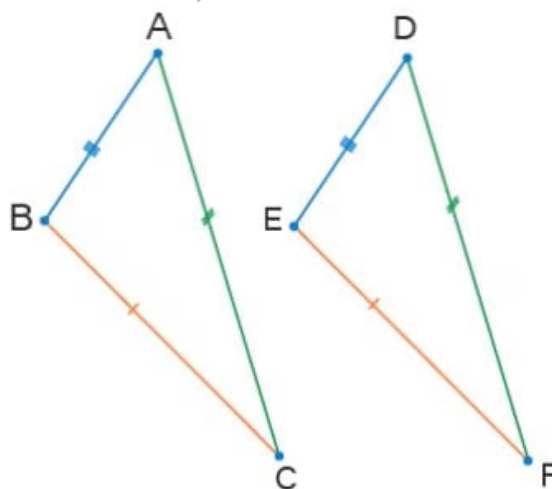
- Two equal circles → same radius.
- Two rectangles → same length and breadth.
- Two shapes drawn differently but same dimensions → congruent.

3. Congruence of Triangles

Triangles are the main focus of the chapter. Two triangles are congruent if their corresponding sides and angles are exactly the same. To check congruence, the following conditions are used.

(1) SSS - Side Side Side

If **all three sides** of one triangle are equal to the three sides of another triangle,
→ The triangles are congruent.



$AB = DE$, $BC = EF$, and $AC = DF$. Therefore, we can say that $\triangle ABC \cong \triangle DEF$.

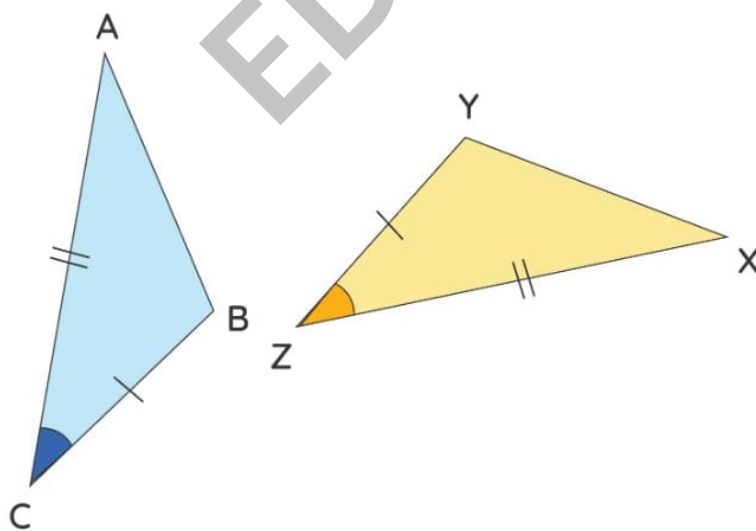
Key idea:

Side lengths alone fix the triangle completely.

(2) SAS - Side Angle Side

If **two sides and the angle between them** are equal,

→ The triangles are congruent.



$AB = XZ$, $BC = YZ$, and $\angle C = \angle Z$. Therefore, $\triangle ACB \cong \triangle XZY$.

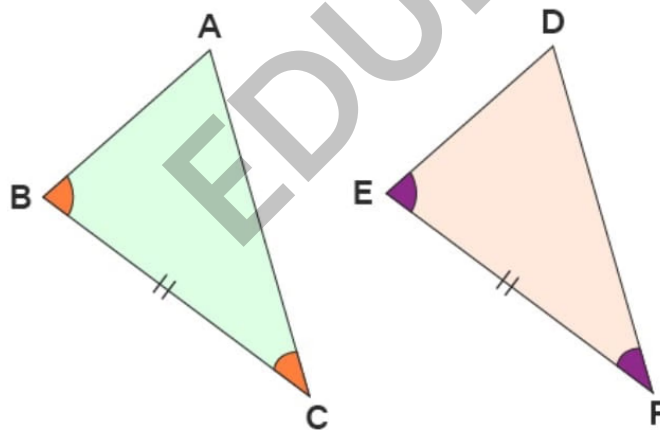
Why it works:

The included angle fixes the opening between the two sides.

(3) ASA - Angle Side Angle

If **two angles and the included side** are equal,

→ The triangles are congruent.



$BC = EF$

$$\angle B = \angle E$$

$$\angle C = \angle F$$

We say that by ASA criterion: $\triangle ABC \cong \triangle DEF$.

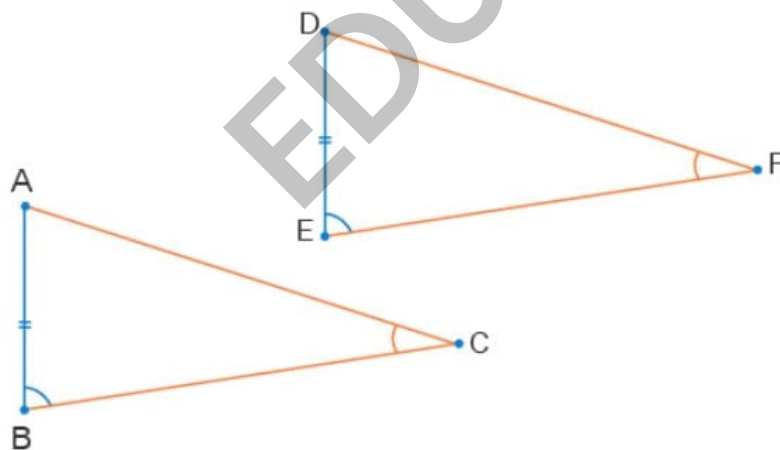
Why it works:

Two angles fix the shape of the triangle, and one side fixes the size.

(4) AAS - Angle Angle Side

If **two angles and any one side** are equal,

→ The triangles are still congruent.



$$AB = DE, \angle B = \angle E, \text{ and } \angle C = \angle F$$

Therefore, according to the ASA congruence rule, $\triangle ABC \cong \triangle DEF$.

Reason:

If two angles are equal, the third angle automatically becomes equal. Then ASA applies.

(5) RHS - Right Hypotenuse Side

(Only for right-angled triangles)

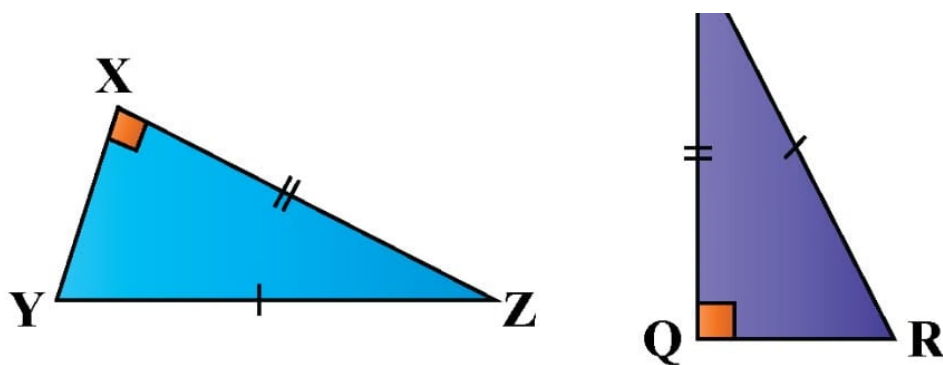
If:

the **right angle**,

the **hypotenuse**, and

one **other side** are equal,





side XZ = side PQ

side YZ = side PR

$\angle ZXY = \angle PQR = 90^\circ$

$\therefore \triangle ZXY \cong \triangle PQR$, by RHS congruence criterion.

→ The triangles are congruent.

Special rule:

This works only in right triangles because the hypotenuse fixes the triangle strongly.

A Condition That Does NOT Ensure Congruence is SSA - Side Side Angle. Two sides and a non-included angle do **not** always give a unique triangle. Two different triangles may be possible.

CPCTC - Corresponding Parts of Congruent Triangles

When two triangles are congruent:

Corresponding **angles** are equal.

Corresponding **sides** are equal.

Corresponding **vertices** match in order.

If:

$$\triangle ABC \cong \triangle XYZ$$

Then:

$$A \leftrightarrow X$$

$$B \leftrightarrow Y$$

$$C \leftrightarrow Z$$

This helps in writing correct congruence statements.

6. Important Properties Derived From Congruence

(1) Angles Opposite Equal Sides Are Equal

In any triangle,

If two sides are equal $\rightarrow$ the angles opposite them are also equal.

Example:

In $\triangle ABC$, if $AB = AC \rightarrow \angle C = \angle B$.

This is the key property of **isosceles triangles**.

(2) Properties of an Equilateral Triangle

In an equilateral triangle:

All sides are equal

All angles are equal

Each angle = **60°**

This follows because all sides are equal $\rightarrow$ all angles opposite them must be equal $\rightarrow$ they sum to 180° .

7. How to Check Congruence for Common Shapes

Circles

Two circles are congruent if:

Their **radii are equal**.

Rectangles

Two rectangles are congruent if:

Their **lengths** and **breadths** are equal.

Summary

Rule	Requirement	Works When
SSS	All 3 sides equal	Always
SAS	Two sides + included angle equal	Always

ASA	Two angles + included side equal	Always
AAS	Two angles + any side equal	Always
RHS	Right angle, hypotenuse, one side equal	Right-angled triangles only
SSA	Two sides + non-included angle	Does NOT guarantee congruence